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• QUANTUM NAVIGATION

PETER DUNNIECE · LORCAN GRIMES · ALEXANDER O'SULLIVAN

EUDIS DEFENCE HACKATHON 2026 · DUBLIN · CHALLENGE 1

Developing a Quantum Enabled sensor platform for GNSS denied navigation

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GNSS jamming is now routine

123,000

Baltic flights hit by GNSS interference in four months

Report to ICAO, Jan–Apr 2025

~1,500

flights a day. It has reached Irish-controlled airspace.

IATA via Irish Times, Jun 2026

+193%

GNSS spoofing, 2023 to 2025. Jamming up 67%.

IATA via Irish Times, Jun 2026

Irish maritime awareness – the jamming problem

Cameras	Lidar, radar	Star trackers	Inertial only	Magnetic map
FAILS	FAILS	FAILS	FAILS	WORKS

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Big platforms already fly MagNav

NORTHROP + SANDBOXAQ · 2026

Quantum MagNav on the Lumberjack uncrewed aircraft, over open water.

HONEYWELL + DIU · SEP 2026

An Embraer 170 flew 4 h 23 min over the Pacific without GPS.

Q-CTRL · 2025

Airborne quantum MagNav with up to ~46× lower error than an aided strategic-grade INS.

Not solved for small drones

On a small drone the sensor sits centimetres from the motors, and the drone's own field swamps the anomaly.

Northrop Grumman, Sep 2026 · DefenseScoop, Sep 2026 · arXiv:2504.08167

Why us?



Peter Dunniece

QUANTUM

- MSc Quantum Physics
- Imperial College London
- Paper In review (Quantum Science and technology)
- Neutral atom Quantum computing
- Beginning PhD Computer Vision



Lorcan Grimes

AEROSPACE

- BEng Aerospace Engineering
- Queen’s University Belfast
- Design Engineer at Boeing
- Graduated top of year
- ESA training, Patch SF Fellowship



Alex O’Sullivan

MECHANICAL

- Mechanical Engineering,
- Trinity College Dublin
- Drone startup background, Dogpatch Labs; Patch SF Fellowship
- Ex Founder

Ireland In Unique Position

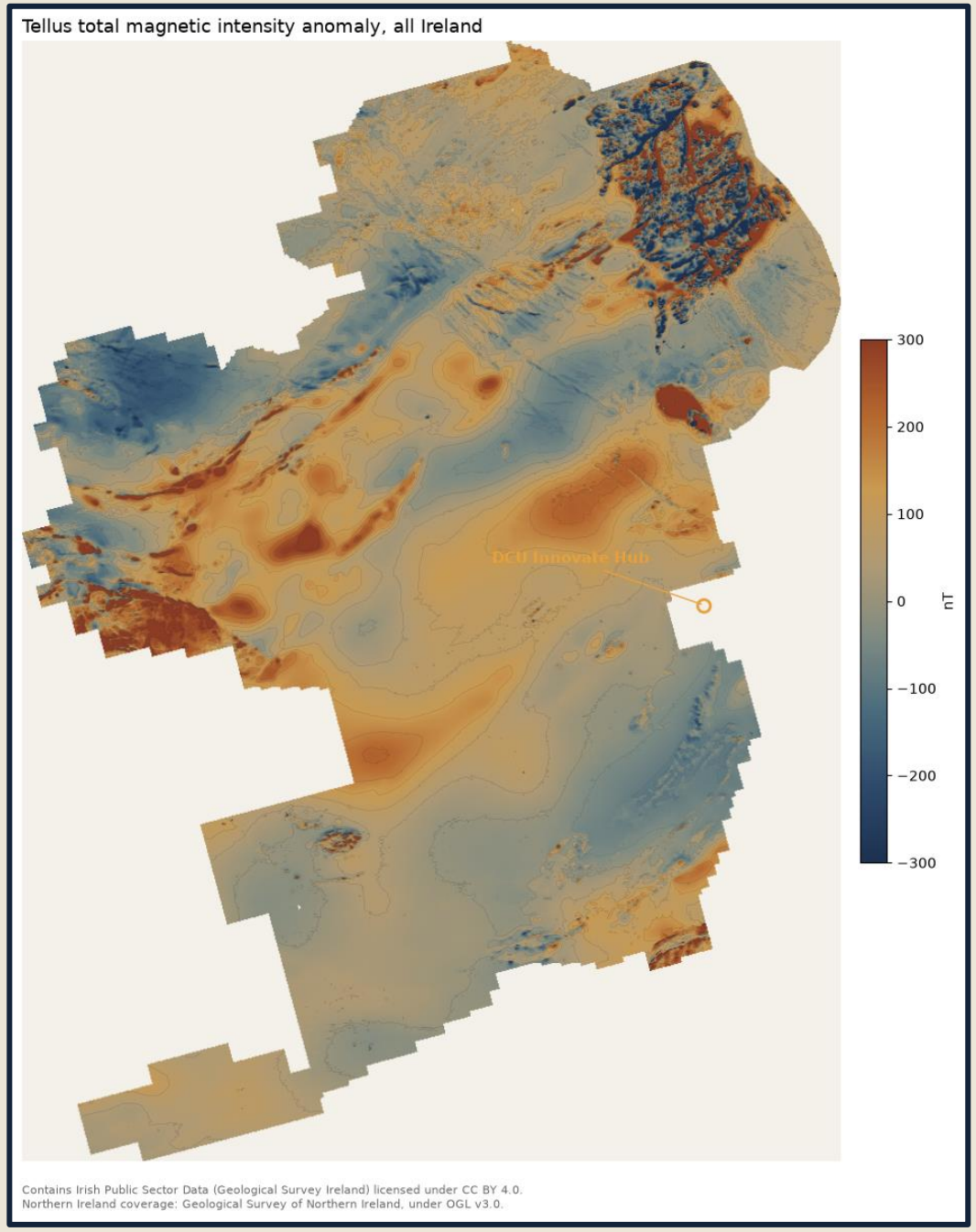
Tellus Mission – 2011 → June 2026

Gathered magnetic measurements from specialised low-flying aircraft.

Dublin, Kerry, Clare release 2027

less than 5% of the United States mapped

Why stop just on land mass?

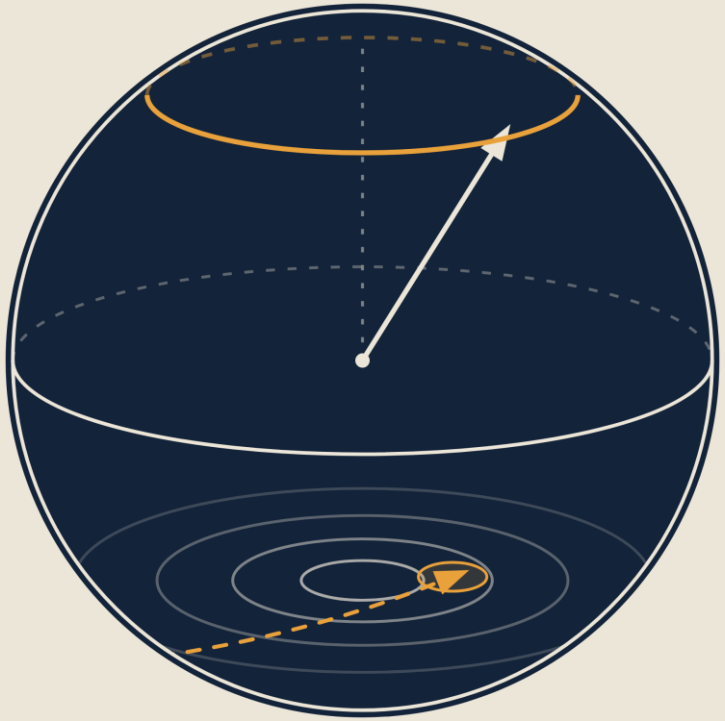


Quantum Scalar OPMs

Circularly polarised 795 nm laser light lines up the spins of rubidium atoms in a heated vapour cell.

In the Earth's field the electron spin wobbles around the field direction.

The light through the cell changes proportional to magnitude of magnetic field strength, doesn't matter what direction the vehicle moves.



The Bloch sphere in our logo is the precessing spin

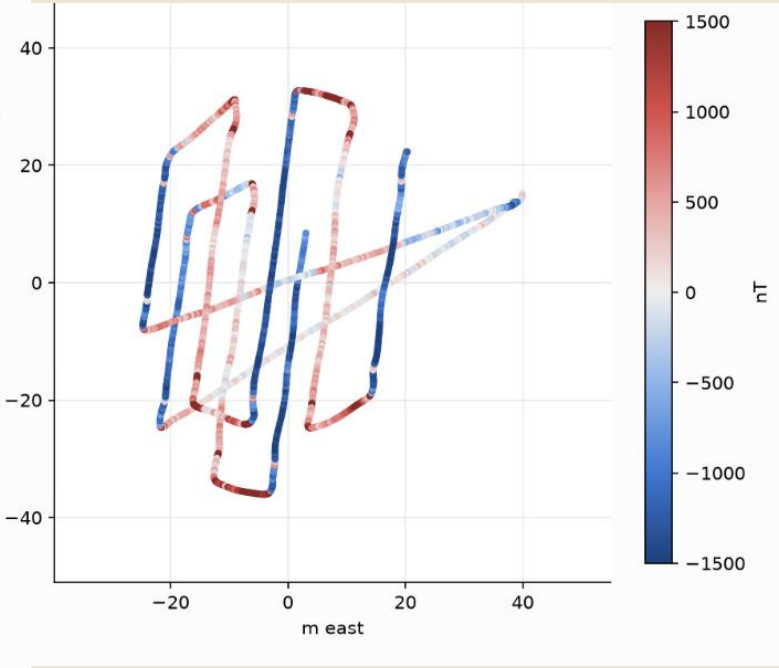
“The European Commission will also initiate a *Quantum Sensing Space and Defence Roadmap* in 2026”

Crashed, rebuilt, flying autonomous surveys



Flying again in [11] hours.





- Fri: two autonomous survey runs at DCU, then return-to-launch and auto-land.
- Every motor's RPM logged next to the magnetometers.

DCU is outside the Tellus survey, so these flights test the platform and the interference model, not map matching.

V1 · Sensor on the frame



TO NEAREST MOTOR

17 cm

sensor to the closest motor

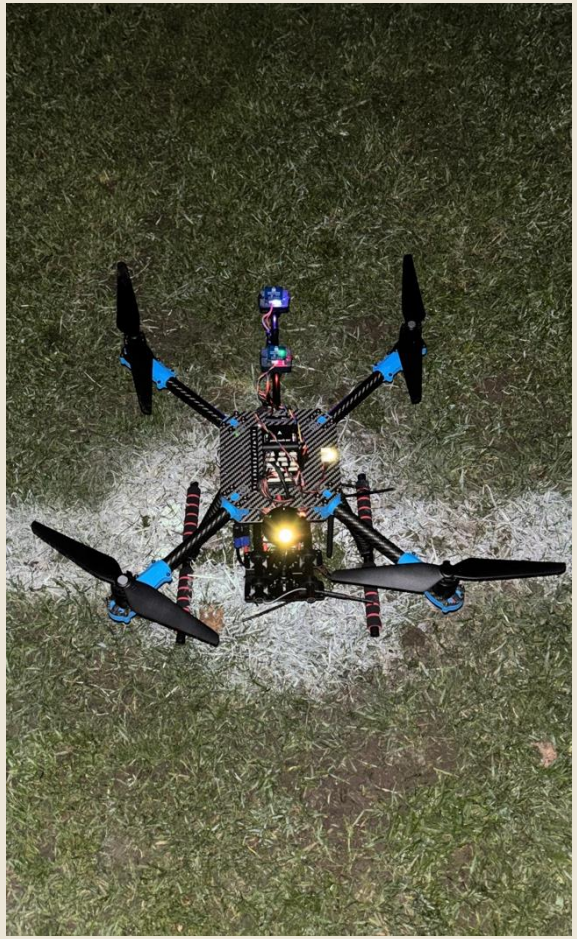
INTERFERENCE

538 nT

against a 28–100 nT map signal

V2 · First boom





TO NEAREST MOTOR

30 cm

sensor to the closest motor

INTERFERENCE

300 nT

against a 28–100 nT map signal

V3 · Longer boom







TO NEAREST MOTOR

70 cm

sensor to the closest motor

INTERFERENCE

153 nT

against a 28–100 nT map signal

GNSS-denied drift, bounded

WITHOUT A MAP

error ≈ t^2

A gyro bias tilts the platform. Gravity leaks into position, and the error grows with the square of time.

WITH COHERA

- 2,000 guesses of how wrong the INS is
- Each scored against the readings every 2 s
- Error budget inflated for correlated map error
- Protection level flags when it’s lost



Over mapped ground, navigating without GNSS becomes a bounded-error problem.

Simulated 10-min survey Tellus, Co. Mayo, measured X500 IMU

We want to develop a flight unit: €30k, built in Ireland

Item	Cost	Note
Airframe and propulsion	~€1,500	printed fixed wing
Autopilot, GNSS, radio	~€1,000	Pixhawk-class, ArduPilot
Quantum magnetometer	~€7,000	QuSpin QTFM class
Navigation computer	~€500	runs the filter
IMU	~€3,000	tactical-grade INS
Integration, calibration, test	~€2,000	
Total	**~€14,100**	

FOR SCALE

~€100k+

a navigation-grade INS

BUILT IN IRELAND

Map: Tellus, open data
Navigation software: ours
Airframe and assembly: made here
One import: the quantum sensor (US) – can change

Indicative estimates, ±50%

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